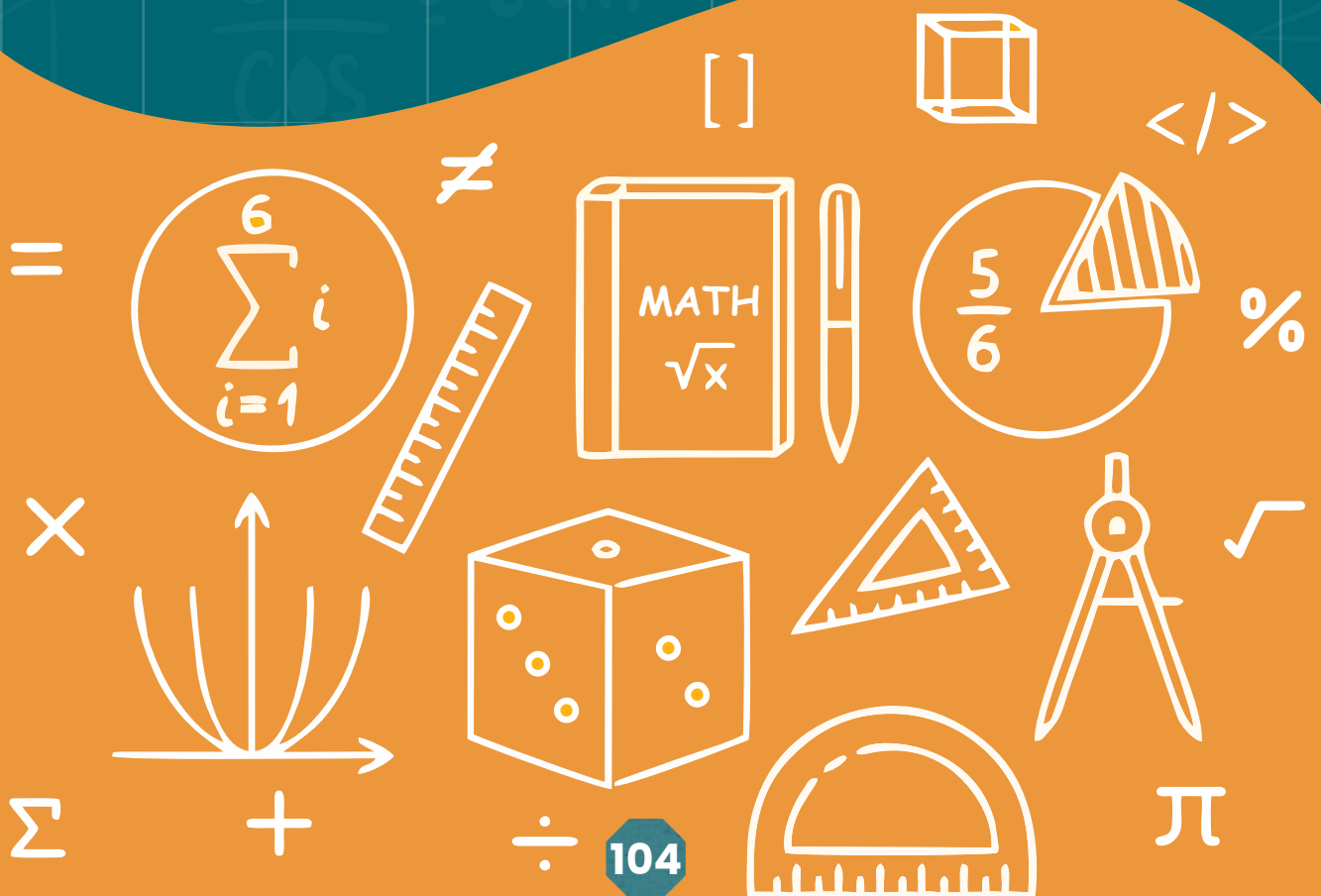


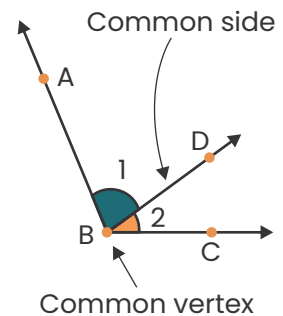
UNIT 10

FUNDAMENTALS OF GEOMETRY



ADJACENT ANGLES

Adjacent angles are those angles that are always placed next to each other in such a way that they share a common vertex and a common side but they do not overlap each other. Observe the following figure to understand what adjacent angles look like. Angle 1 and 2 are adjacent because they have a common side BD and a common vertex B.



COMPLEMENTARY ANGLES

When the sum of two angles is 90° , then the angles are known as complementary angles. In other words, if two angles add up to form a right angle, then these angles are referred to as complementary angles. Here we say that the two angles complement each other.

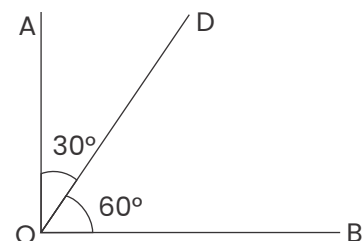
How to Find Complementary Angles?

Suppose if one angle is x then the other angle will be $90^\circ - x$. Hence, we use these complementary angles for trigonometry ratios, where one ratio is complementary to another ratio by 90 degrees such as;

- $\sin (90^\circ - A) = \cos A$ and $\cos (90^\circ - A) = \sin A$
- $\tan (90^\circ - A) = \cot A$ and $\cot (90^\circ - A) = \tan A$
- $\sec (90^\circ - A) = \operatorname{cosec} A$ and $\operatorname{cosec} (90^\circ - A) = \sec A$

Hence, you can see here the trigonometric ratio of the angles gets changed if they complement each other.

In the given figure, the measure of angle BOD is 60° and angle AOD measures 30° . By adding both of these angles we get a right angle, therefore $\angle BOD$ and $\angle AOD$ are complementary angles.



SUPPLEMENTARY ANGLES

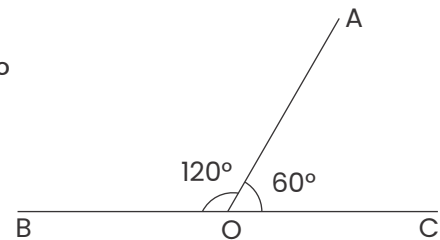
When the sum of two angles is 180° , then the angles are known as supplementary angles. In other words, if two angles add up, to form a straight angle, then those angles are referred to as supplementary angles.

How to Find Supplementary Angles?

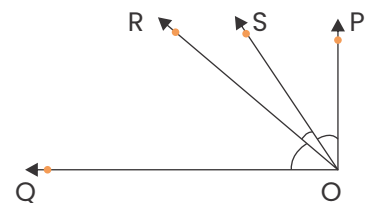
The two angles form a linear angle, such that, if one angle is x , then the other the angle is $180^\circ - x$. The linearity here proves that the properties of the angles remain the same. Take the examples of trigonometric ratios such as;

- $\sin (180^\circ - A) = \sin A$
- $\cos (180^\circ - A) = -\cos A$ (quadrant is changed)
- $\tan (180^\circ - A) = -\tan A$

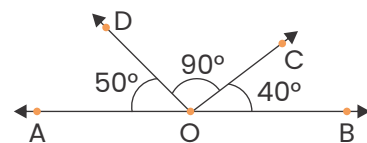
In given figure, the measure of $\angle AOC$ is 60° and $\angle AOB$ measures 120° . By adding both of these angles we get a straight angle. Therefore, $\angle AOC$ and $\angle AOB$ are supplementary angles, and both of these angles are known as a supplement to each other.

**Exercise**

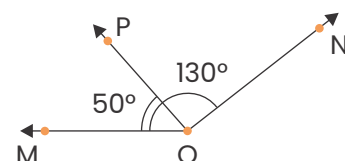
1. Write the names of all the adjacent angles shown in **Fig. 1**.



2. Write pairs of all the supplementary angles from **Fig. 2**.



3. In Fig. 3, $m\angle MON = 130^\circ$ and $m\angle MOP = 50^\circ$. Find the measure of the adjacent angle $\angle PON$



EXERCISE 10.1

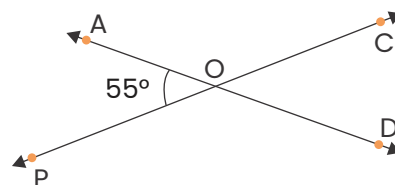
1. Write the measures of the complement of the following.

Measure of the Given Angle	49°	62°	54°	36°	67°	81°	15°
Measure of its Complement							

2. Write the measures of the supplement of the following supplement angles.

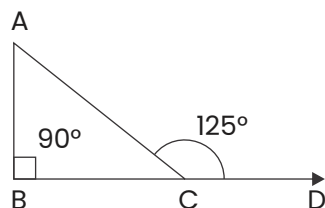
Measure of the Given Angle	80°	132°	49°	76°	125°	103°	68°
Measure of its Supplement							

3. Look at **Fig. 4** and write the measures of all pairs of vertically opposite angles.

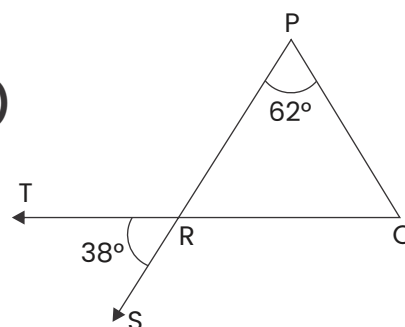


4. Find the measurement of the unknown angles in the following triangles:

(i)



(ii)



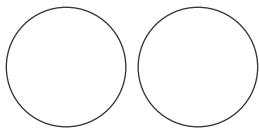
CONGRUENT AND SIMILAR FIGURES

Identify congruent and similar figures:

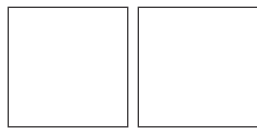
Congruent Figures:

That have exactly the same size and same shape are called congruent figures, they need not be identical. They may have different colours or textures.

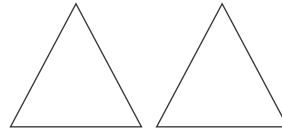
Example: All the squares, rectangles, circles and equilateral triangles having exactly the same size and same shape are congruent figures.



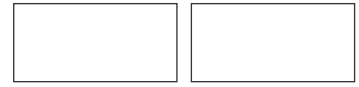
Circles



Squares



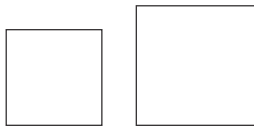
Triangles



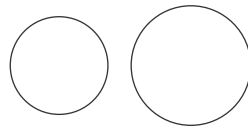
Rectangles

Similar Figures: The figures that have exactly the same shape but not necessarily the same size are called similar figures.

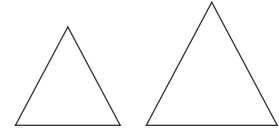
Example: All squares, circles and equilateral triangles are examples of similar figures.



Squares



Circles



Equilateral Triangles

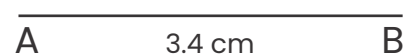
Recognize the symbol of Congruency:

(i) Congruent Figures: Two or more geometrical figures are said to be congruent if these have exactly the same shape and same size. The symbol ' $\cong$ ' stands for 'is congruent to'.

For example:

(i) Two line segments are congruent if their lengths are the same.

Here, $m\overline{AB} = 3.4\text{m} = m\overline{CD}$.



Both have exactly the same shape and same size.

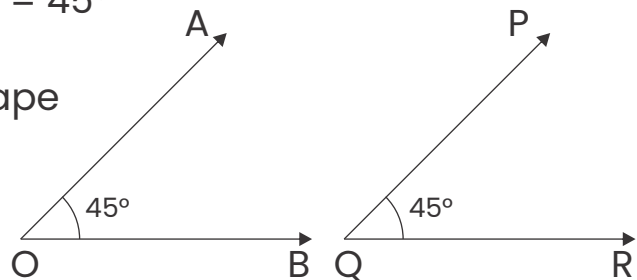
Then $m\overline{AB} = m\overline{CD}$ and $\overline{AB} \cong \overline{CD}$.



(ii) Now Consider Angles: Angles $\angle AOB$ and $\angle PQR$

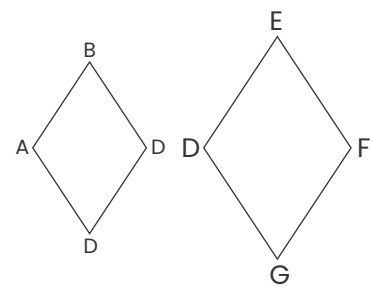
Here, $m\angle AOB = 45^\circ$ and $m\angle PQR = 45^\circ$

Both have exactly the same shape and same size.



Similar Figures: Two or more geometrical figures are said to be similar if their shape is same.

The similarity of geometrical figures is represented by the symbol " $\sim$ ".

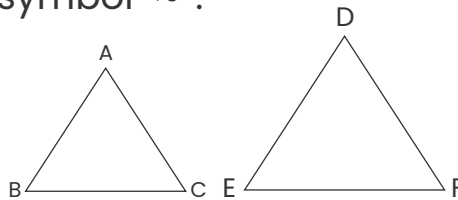


For example:

Quad. ABCD $\sim$ Quad. DEFG

A ——— B

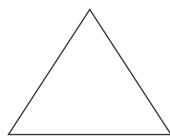
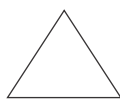
C ——— D



EXERCISE 10.2

1. Write Congruent or Similar Figures in the Following:

i)



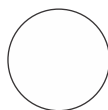
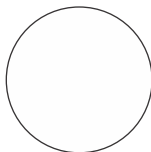
ii)



iii)



iv)

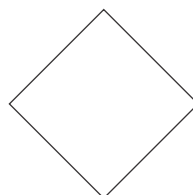
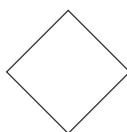


v)

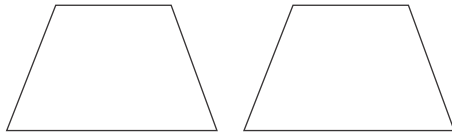
_____ 3 cm

_____ 5 cm

vi)



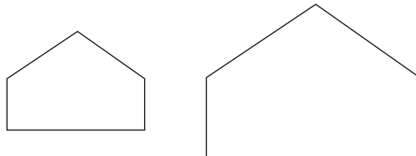
vii)



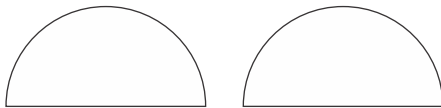
viii)



ix)



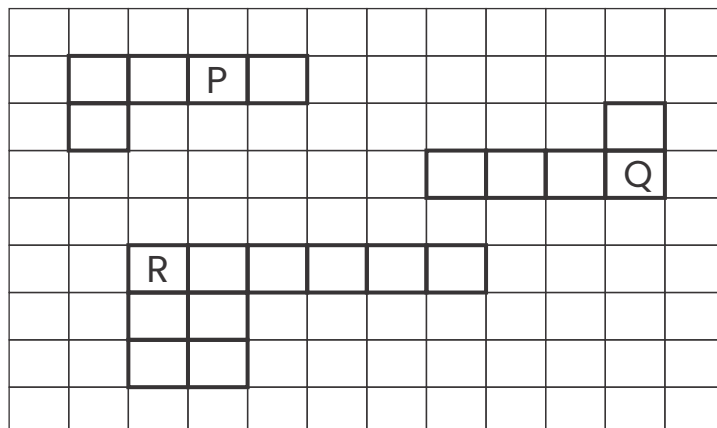
x)



2. Draw the following geometrical shapes:

- (i) Draw two rectangles of different sizes. Are they similar?
- (ii) Draw two circles of different radii. Are they congruent?
- (iii) Draw two equilateral triangles of the same measures. Are they similar?
- (iv) Draw two squares of different sizes. Are they congruent?
- (v) Draw two rectangles of the same size. Are they congruent?

3. Look at the graph and consider figure P, Q, and R, then answer the following questions.



- (i) Has figure P the same shape as figure R?
- (ii) Has figure Q the same size as figure R?
- (iii) Are P and Q congruent?.

Apply the properties for two figures to be congruent or similar.

We have already discussed the properties of congruent and similar figures:

1. Congruent figures have the same shape and same size.
2. Similar figures have the same shape (i.e., same angles and the same ratio of sides) but not necessarily the same size.

Now, we use the above two properties to classify figures as congruent or similar.

Example 1:

Look at the pair of figures and identify whether they are congruent or similar.

(i) Line Segments

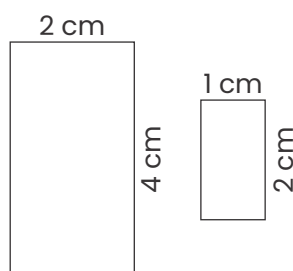
- Both line segments have the same shape and same size.
- Therefore, $PQ \cong RS$ (They are congruent).

P 2.7 cm Q

R 2.7 cm S

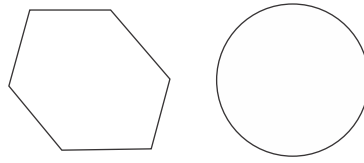
(ii) Rectangles

- The rectangles have the same shape but different sizes.
- They also maintain the same ratio in the measure of lengths of the sides.
- Therefore, these are similar figures.

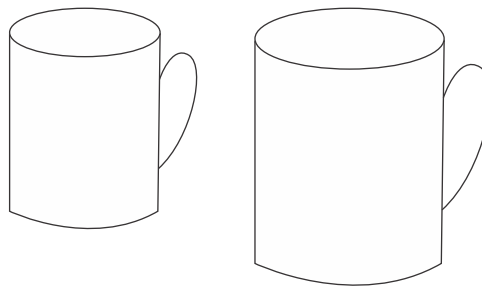


(iii) Different Shapes

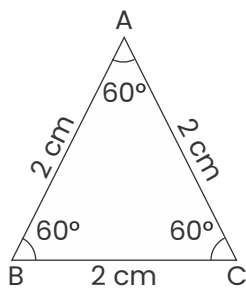
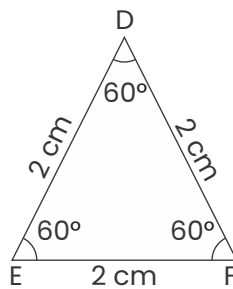
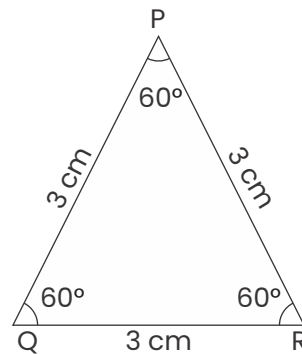
- The given figures are of different shapes.
- Therefore, these figures are neither similar nor congruent.

**(iv) 3D Objects**

- Both figures represent the same type of object.
- They have almost the same shape but different sizes.
- Therefore, these are similar figures.

**Example 2:**

Identify and Write Similar and Congruent Triangles. Let us observe the following triangles:

**Fig (i)****Fig (ii)****Fig (iii)**

In $\triangle ABC$ and $\triangle DEF$,

all the corresponding sides and angles are equal.

$$m\overline{AB} = m\overline{DE}, m\overline{BC} = m\overline{EF} \text{ and } m\overline{CA} = m\overline{FD}$$

$$m\angle ACB = m\angle DFE,$$

$$m\angle ABC = m\angle DEF,$$

$$m\angle BAC = m\angle EDF$$

Therefore, $\triangle ABC \cong \triangle DEF$ read as $\triangle ABC$ is congruent to $\triangle DEF$.

Consider Fig (ii) and (iii) DEF is the original triangle and PQR is the triangle enlarged.

Therefore, DEF is similar to PQR.

Here the corresponding angles are the same and the length of each side is increased by the same ratio, i.e., the corresponding sides are in the same proportion:

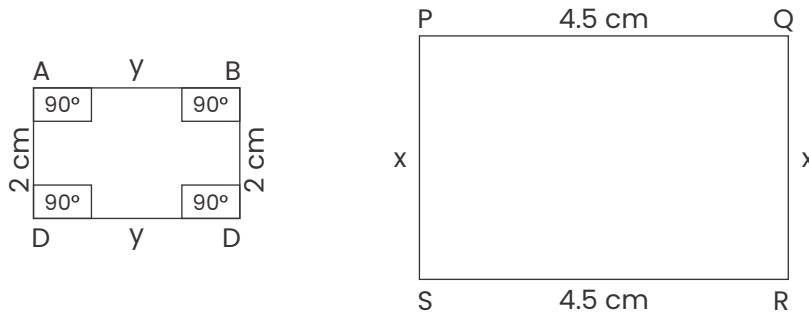
$$\frac{\overline{DE}}{\overline{PQ}} = \frac{\overline{EF}}{\overline{QR}} = \frac{\overline{FD}}{\overline{RP}} = \frac{2}{3}$$

Hence, DEF ~ PQR

Read as DEF is similar to PQR.

Example 3:

If both rectangles are similar to one another, find the measure of unknowns, when the ratio in the measure of sides is $\frac{3}{2}$.

**Solution:**

Since Rectangle $ABCD \sim$ Rectangle $PQRS$, then:

- (i) Corresponding sides are proportional.
- (ii) Corresponding angles are equal.

$m\angle P = m\angle A = 90^\circ$, $m\angle Q = m\angle B = 90^\circ$, $m\angle R = m\angle C = 90^\circ$, $m\angle S = m\angle D = 90^\circ$

Also $\frac{m\overline{PQ}}{m\overline{AB}} = \frac{4.5}{y} = \frac{3}{2}$ (Corresponding ratio in sides)

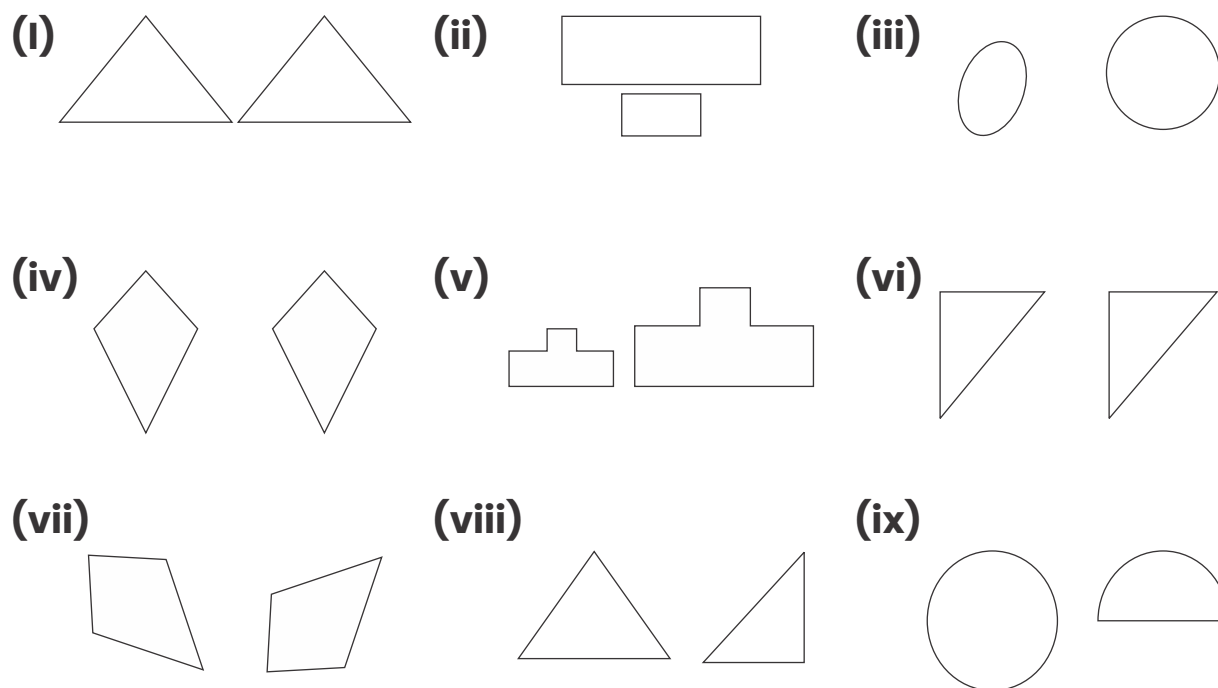
Therefore $y = \frac{(4.5) \times 2}{3} = \frac{9}{3} = 3$, Hence $m\overline{AB} = 3$ cm

Again $\frac{m\overline{QR}}{m\overline{BC}} = \frac{x}{2} = \frac{3}{2}$

Therefore $x = \frac{2 \times 3}{2}$, then $x = 3$ or $m\overline{QR} = 3$ cm.

EXERCISE 10.3

1. Decide whether the following figures, given in each pair are congruent or similar or neither similar nor congruent.



2. Write in symbols the congruent segments from the following:

$m\overline{AB} = 3.2 \text{ cm}$	$m\overline{CD} = 4.5 \text{ cm}$
$m\overline{DE} = 5.1 \text{ cm}$	$m\overline{GH} = 4.8 \text{ cm}$
$m\overline{MN} = 4.5 \text{ cm}$	$m\overline{XZ} = 5.1 \text{ cm}$
$m\overline{YZ} = 4.8 \text{ cm}$	$m\overline{PQ} = 5.4 \text{ cm}$
$m\overline{RS} = 5.4 \text{ cm}$	$m\overline{EF} = 3.9 \text{ cm}$
$m\overline{YZ} = 3.9 \text{ cm}$	$m\overline{XY} = 3.2 \text{ cm}$

3. Write in symbols the congruent angles from the following:

$m\angle ABC = 60^\circ$	$m\angle DEF = 70^\circ$
$m\angle XYZ = 110^\circ$	$m\angle PQR = 45^\circ$
$m\angle COD = 45^\circ$	$m\angle PRQ = 60^\circ$
$m\angle AOB = 70^\circ$	$m\angle MNS = 110^\circ$

4. Fill in the blanks:

(i) $m\angle ABC = 100^\circ$, $\triangle ABC \cong \triangle DEF$, Therefore $m\angle DEF =$ _____

(ii) $m\angle PQR = m\angle XYZ$ Therefore $\angle PQR$ _____ $\angle XYZ$

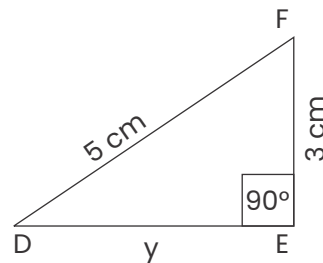
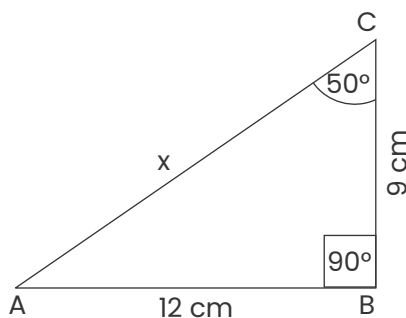
(iii) Line segments having equal lengths are called _____ segments.

(iv) Angles having _____ measures are called congruent angles.

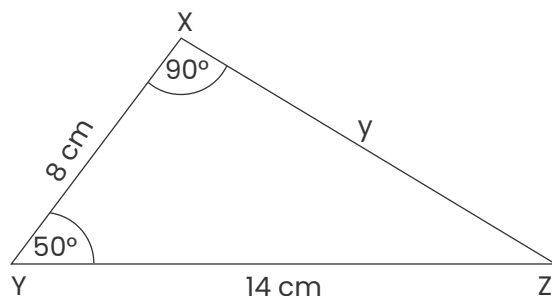
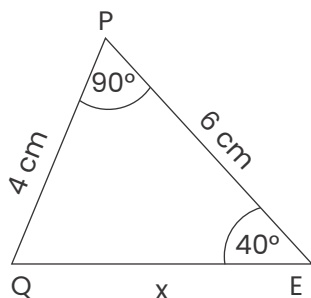
(v) Two circles are congruent if their _____ are equal.

5. Consider the following pair of similar figures. Find the unknown elements x and y in the following:

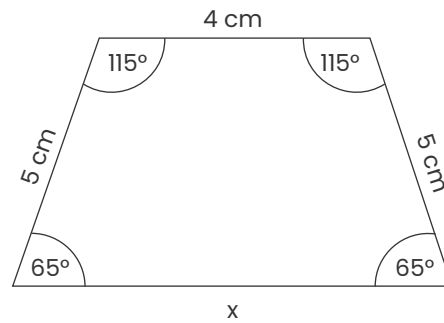
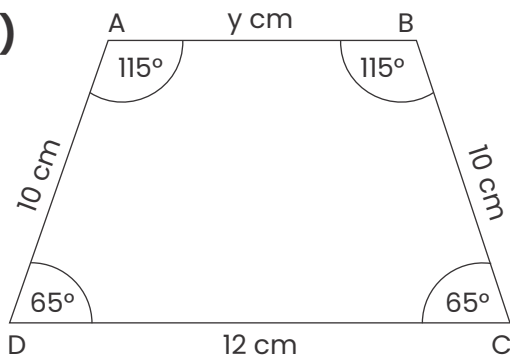
i)



ii)



iii)



CONGRUENT TRIANGLES

Apply the following properties for congruency between two triangles:

1. $SSS \cong SSS$
2. $SAS \cong SAS$
3. $ASA \cong ASA$
4. $HS \cong HS$

These are the four properties for proving the congruency of two triangles.

First Property for Congruency of Triangles ($SSS \cong SSS$)

This property states that if all three sides of one triangle are congruent to the three corresponding sides of another triangle, then both triangles are congruent.

Example: Consider two triangles PQR and XYZ.

$$\triangle PQR \longleftrightarrow \triangle XYZ$$

$\triangle PQR$ has one-to-one correspondence with $\triangle XYZ$, as follows:

(i) $m\overline{PQ} = 2.5 \text{ cm}$, $m\overline{XY} = 2.5 \text{ cm}$

Therefore, $\overline{PQ} \cong \overline{XY}$

(ii) $m\overline{PR} = 3.0 \text{ cm}$, $m\overline{XZ} = 3.0 \text{ cm}$

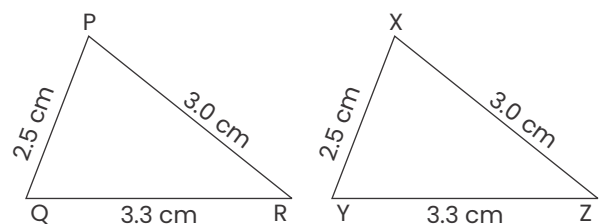
Therefore, $\overline{PR} \cong \overline{XZ}$

(iii) $m\overline{QR} = 3.3 \text{ cm}$, $m\overline{YZ} = 3.3 \text{ cm}$

Therefore, $\overline{QR} \cong \overline{YZ}$

Hence, $\triangle PQR \cong \triangle XYZ$

($SSS \cong SSS$)



Second Property for Congruency of Triangles ($SAS \cong SAS$)

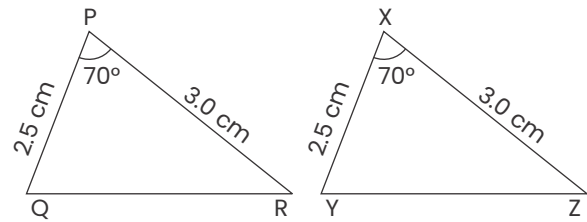
This property states that if two sides and the included angle of one triangle are congruent to the corresponding two sides and the included angle of another triangle, then both triangles are congruent.

Example:

Consider two triangles PQR and XYZ, where:

$$\triangle PQR \longleftrightarrow \triangle XYZ$$

$\triangle PQR$ has one-to-one correspondence with $\triangle XYZ$ as under:

**Third Property for Congruency of Triangles ($ASA \cong ASA$)**

If two angles and the included side of one triangle are equal to the corresponding two angles and the included side of another triangle, the triangles are congruent.

Example:

Consider two triangles PQR and XYZ

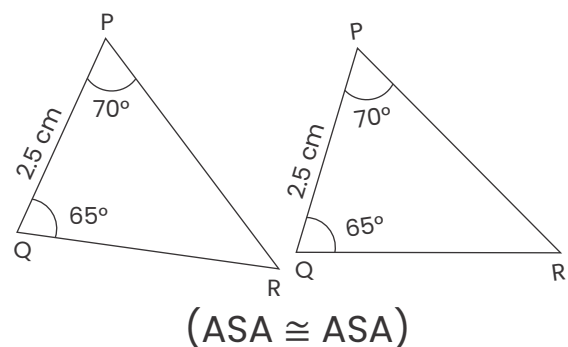
$$\triangle PQR \longleftrightarrow \triangle XYZ$$

$\triangle PQR$ has one-to-one correspondence with $\triangle XYZ$, as follows:

(i) $m\angle PQR = 65^\circ$ and $m\angle XYZ = 65^\circ$
Therefore, $\angle PQR \cong \angle XYZ$

(ii) $m\overline{PQ} = 2.5 \text{ cm}$, $m\overline{XY} = 2.5 \text{ cm}$
Therefore, $\overline{PQ} \cong \overline{XY}$

(iii) $m\angle QPR = 70^\circ$ and $m\angle YXZ = 70^\circ$
Therefore, $\angle QPR \cong \angle YXZ$



Hence, $\triangle PQR \cong \triangle XYZ$

Fourth Property for Congruency of Triangles ($HS \cong HS$)

If the Hypotenuse and one side (Base or Altitude) of a right-angled triangle are congruent to the corresponding Hypotenuse and one side of another right-angled triangle, then the triangles are congruent.

Example:

Consider two right-angled triangles PQR and XYZ

$$\triangle PQR \longleftrightarrow \triangle XYZ$$

$\triangle PQR$ has one-to-one correspondence with $\triangle XYZ$ as follows:

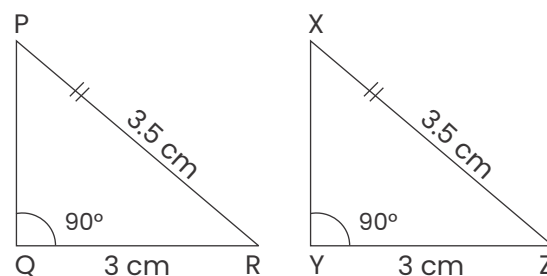
(i) $m\angle PQR = 90^\circ$, $m\angle XYZ = 90^\circ$ i.e. $m\angle PQR = m\angle XYZ = \text{right angle}$.

(ii) $m\overline{QR} = 3 \text{ cm}$, $m\overline{YZ} = 3 \text{ cm}$

Therefore: $\overline{QR} \cong \overline{YZ}$

(iii) $m\overline{PR} = 3.5 \text{ cm}$, $m\overline{XZ} = 3.5 \text{ cm}$

Therefore: $\overline{PR} \cong \overline{XZ}$



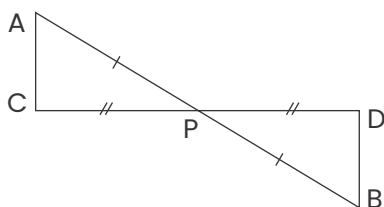
Then $\triangle PQR \cong \triangle XYZ$

($HS \cong HS$)

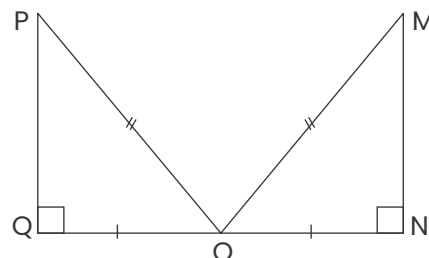
EXERCISE 10.3

1. In the figures below, the congruent parts of triangles have been indicated by like marking. Write the congruent parts in order and state by which property the triangles are congruent.

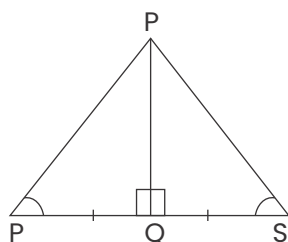
(i)



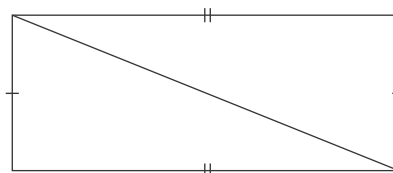
(ii)

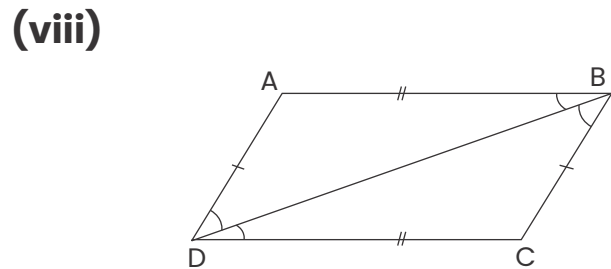
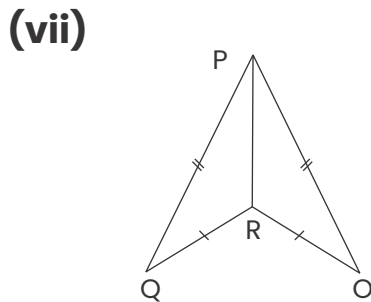
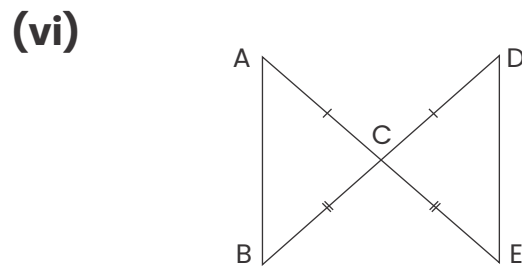
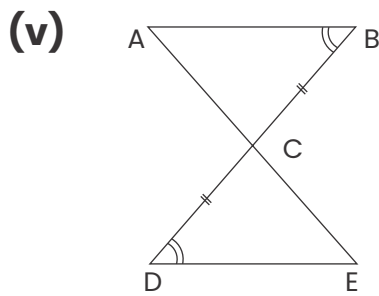


(iii)



(iv)

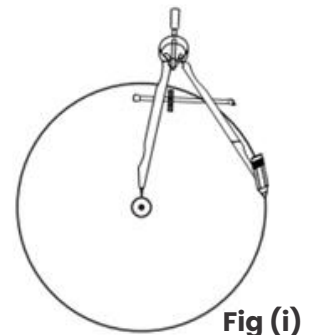




CIRCLE

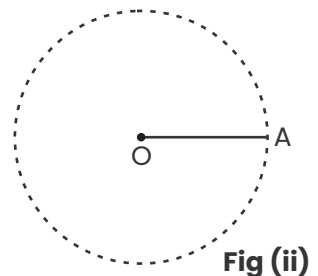
i. A circle and its centre:

To draw a circle and its centre, start by marking a point O on a paper. Place the sharp end of a compass on O and open the compass to your desired radius. Rotate the pencil around O while keeping the sharp end steady. The shape you create is a circle, and the point O is its centre.



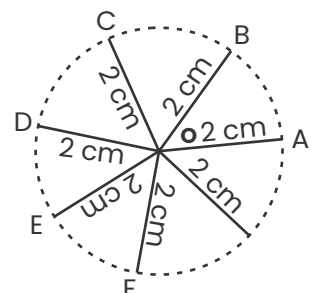
ii. Radius of a circle:

The radius of a circle is the distance from the center of the circle to any point on its boundary. It is also the distance between the two ends of a compass when drawing the circle.



The term "Radius of a circle" is used to denote a distance. For example, OA is the radial segment of the circle, or the radius is 2 cm.

Remember: A circle has an infinite number of radial segments. In Fig. (iii), all these radii are equal in length. i.e. $m\overline{OA} = m\overline{OB} = m\overline{OC} = m\overline{OD} = m\overline{OE} = m\overline{OF} = m\overline{OG} = 2 \text{ cm}$.



iii. Diameter

A line segment joining any two points on the circle, passing through the center of the circle is called the Diameter of the circle. Diameter is also used to denote a line segment as well as a distance.

Remember: The diameter of a circle passes through its centre. Its length is twice the length of the radius.

A circle has an infinite number of diameters, and all these diameters are equal in length. In Fig. (v): $\overline{mAB} = \overline{mCD} = \overline{mEE} = \overline{mGH}$

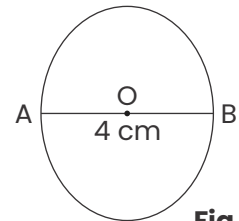


Fig (iv)

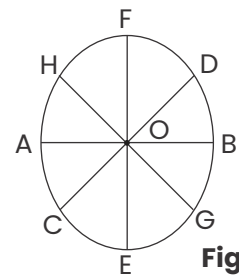


Fig (v)

iv. Chord

A line segment joining any two points on the circle is called a chord of the circle.

Remember: A circle has an infinite number of chords. Every diameter is the longest chord of the circle.

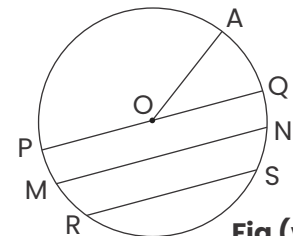


Fig (vi)

v. Arc

Consider a circle. In Fig. (vii) take any two points A and B on the circle. These two points divide the circle into two parts. Each of these parts is called an arc of the circle. We have two types of arcs.

(a) Minor Arc

(b) Major Arc.

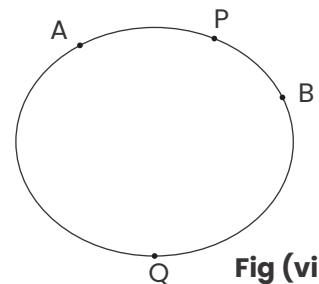


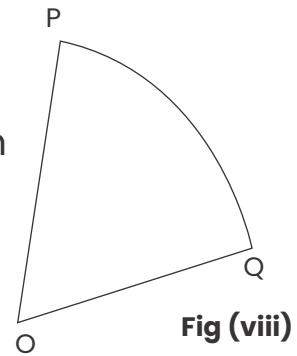
Fig (vii)

Minor Arc: An arc which is smaller than half of the circle is called a minor arc. In the above figure, APB is a minor arc or AB is a minor arc. A minor arc is named with two endpoints.

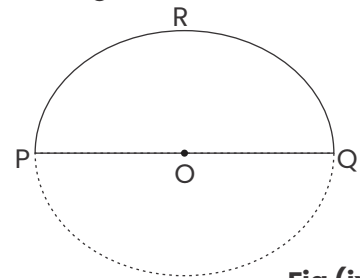
Major Arc: An arc which is more than half of the circle is a major arc. In the above figure, AQB is a major arc.

vi. Sector:

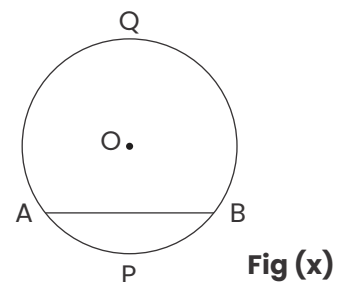
A sector is a part of a circle that looks like a slice of pizza. It is formed when a circular region is divided, creating a curved shape. The two straight sides of the sector meet at the center of the circle.

**Fig (viii)****vii. Semi-Circle:**

Semi-circle means half of a circle. In the adjoining figure, there is a semi-circle with PRQ.

**Fig (ix)****viii. Segment of a circle:**

The segment of a circle is a shape whose boundary is an arc of the circle and a chord of the circle.

**Fig (x)****EXERCISE 10.4**

Fill in the blanks with proper words.

1. The segment joining any two points on the circle is called the _____ of the circle.
2. The distance between the centre of a circle and any point on the circle is called the _____ of the circle.
3. The segment joining the centre of a circle to any point on the circle is called the _____ of the circle.
4. All the points on a circle are equidistant from the _____.
5. A chord passing through the centre of a circle is called the _____ of the circle.
6. The diameter of a circle passes through its _____ and its

length is _____ of the length of the radius.

7. The longest chord of a circle is the _____ of the circle.

8. There are _____ number of radial segments of any given circle and they are _____ in length.

9. The radius of a circle is _____ of the diameter of the circle.

10. There are _____ number of diameters of any given circle and they are _____ in length.

Demonstrate the property, all the angles inscribed in the semi-circle are right angles.

Let's demonstrate: "The angle in a semi-circle is a right angle."

i. - Draw a semi-circle with a suitable radius.

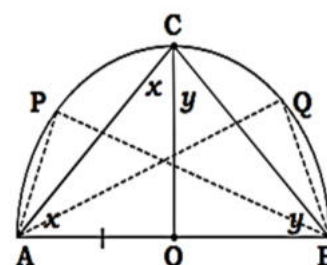
- Mark its center as O.

- Draw a diameter AB through the center O.

ii. - Make the diameter as one side of a triangle.

- Draw the other two sides from points A and B to meet at a vertex C on the circumference.

Thus, we get a $\triangle ABC$.



Now measure $\angle ACB$ with the protractor; we will see $m\angle ACB = 90^\circ$ i.e. $\angle ACB$ is a right angle.

iii. Again take two points P and Q on the semi-circle. Draw triangles APB and AQB.

iv. Measure $\angle APB$ and $\angle AQB$ with the protractor. We will see that: $m\angle APB = m\angle AQB = 90^\circ$.

Thus, the angle in a semi-circle is a right angle.

Hence proved that the angle in a semi-circle is a right angle.

Draw a segment of a circle and demonstrate the property, the angles in the same segment of a circle are equal.

Let us demonstrate the property: "The angles in the same segment of a circle are equal."

Steps of Demonstration:

i. Draw a circle with a suitable radius.

ii. Draw a chord AB of the circle, other than a diameter.

iii. We have two segments: (i) Minor (ii) Major.

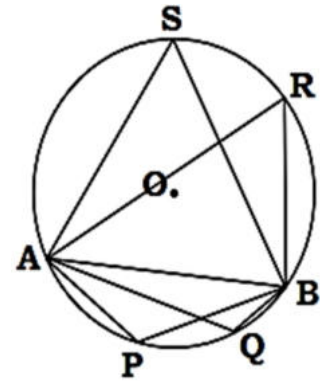
iv. In the minor segment of the circle, draw two inscribed angles of the smaller segment $\angle APB$ and $\angle AQB$ with a scale.

v. Now measure them with the protractor, we will see that:
 $m\angle APB = m\angle AQB$

vi. Again draw two angles in the major segment of the circle, $\angle ASB$ and $\angle ARB$, with a scale.

vii. Measure them with the protractor, we will see that:
 $m\angle ASB = m\angle ARB$

It is concluded that the angles in the same segment of a circle are equal.



EXERCISE 10.5

- 1.** Draw two semi-circles of different radial segments say, 2.5 cm and 3 cm. Draw an angle in every semi-circle and verify it is a right angle.
- 2.** Draw a semi-circle with radius 3.3 cm. Demonstrate the property of a semi-circle. "The angle in a semi-circle is of measure 90° ."
- 3.** Draw a circle of radius 2.7 cm, demonstrate the property: "The angles in the same segment of a circle are equal."
- 4.** Draw a circle of radius 2.8 cm. Draw the angles in the same segment of a circle. Verify that these angles are equal in measure.